

Fluid Pressure and Hydrostatic Pressure

Dr. Huan Phan
Assistant Professor, Robotics and Mechatronics Engineering
Randolph College

Fluid Mechanics Lecture Notes

Learning Objectives

By the end of this lecture, students should be able to:

- Explain why pressure at a point in a fluid acts equally in all directions.
- Apply the hydrostatic pressure equation to fluids at rest.
- Distinguish between absolute pressure and gauge pressure.
- Determine when pressures are equal in a continuous fluid.
- Solve basic manometer pressure problems.

1 Introduction

One of the most important concepts in fluid mechanics is **fluid pressure**. Together with viscosity, pressure is a fundamental property that helps us understand how fluids behave, especially when the fluid is at rest.

This topic is important for solving problems involving hydrostatic pressure fields, reservoirs, tanks, and manometers.

2 Pressure Acts Equally in All Directions

The first important property of fluid pressure is that **pressure at a point in a fluid acts equally in all directions**.

Imagine a small fluid element inside a fluid. No matter what direction we look at, the pressure at that same point has the same magnitude. This is true whether the pressure acts horizontally, vertically, or along a slanted surface.

To understand this, consider a triangular fluid element. We use a triangular shape because it allows us to compare pressure forces in the horizontal, vertical, and inclined directions at the same time.

Suppose the pressure acting on the vertical face is P_y , the pressure acting on the horizontal face is P_z , and the pressure acting on the slanted face is P .

Assume the fluid is not accelerating. Therefore, the sum of forces in each direction must be zero. If we take the sum of forces in the y -direction, the horizontal pressure force from P_y must balance the horizontal component of the pressure force from the slanted surface.

After simplifying the force balance, we find that

$$P_y = P. \quad (1)$$

If we do the same force balance in the z -direction, we also find that

$$P_z = P. \quad (2)$$

Therefore,

$$P_y = P_z = P. \quad (3)$$

This means that pressure at a point in a fluid is the same in every direction. This principle is known as **Pascal's Law**.

3 Coordinate System in Fluid Mechanics

In many engineering courses, such as statics or mechanics of materials, the coordinate system may be drawn with the x , y , and z axes in a familiar orientation.

However, in fluid mechanics, we usually use the z -axis to represent height. This means that the x - and y -axes are usually placed in the horizontal plane, while the z -axis points upward.

The exact coordinate system is not the most important thing. What matters is that we stay consistent. In this lecture, we will use z as the vertical direction, pointing upward.

4 Hydrostatic Pressure Increases with Depth

The second important concept is that **hydrostatic pressure increases with depth**.

The word “hydrostatic” does not only mean water. It refers to a fluid at rest. Therefore, hydrostatic pressure means pressure in a stationary fluid.

Consider a very small fluid element with dimensions dx , dy , and dz . Let the pressure at the center of the element be P . The pressure on each side of the element can be written in terms of P and the pressure change in that direction.

For example, the pressure on one side in the y -direction can be written as

$$P + \frac{\partial P}{\partial y} \frac{dy}{2}, \quad (4)$$

and on the opposite side as

$$P - \frac{\partial P}{\partial y} \frac{dy}{2}. \quad (5)$$

If we apply force balance in the y -direction, the pressure forces must balance because the fluid is not accelerating. This gives

$$\frac{\partial P}{\partial y} = 0. \quad (6)$$

Similarly, in the x -direction,

$$\frac{\partial P}{\partial x} = 0. \quad (7)$$

This means that pressure does not change horizontally in a fluid at rest.

However, in the vertical z -direction, we must include the weight of the fluid element. The weight of the small element is

$$\gamma \, dx \, dy \, dz, \quad (8)$$

where γ is the specific weight of the fluid.

When we apply force balance in the vertical direction, we obtain

$$\frac{\partial P}{\partial z} = -\gamma. \quad (9)$$

This equation tells us that pressure decreases as z increases upward. In other words, pressure increases as we move downward into the fluid.

Key Idea

The deeper we go in a static fluid, the larger the pressure becomes.

5 Pressure Is the Same at the Same Height in a Continuous Fluid

The third important concept is that **pressure is the same at the same elevation in the same continuous fluid**.

If two points are located at the same height in the same connected fluid, then the pressures at those two points are equal.

For example, if we move horizontally through the same fluid without crossing a solid wall or changing to another fluid, the pressure remains the same. Pressure only changes when we move up or down.

This makes sense physically. If two points at the same height in the same fluid had different pressures, the fluid would experience a net force from the higher-pressure region toward the lower-pressure region. That would cause the fluid to accelerate, which contradicts the assumption of hydrostatic conditions.

Therefore, for a static continuous fluid,

$$P_1 = P_2, \quad (10)$$

if points 1 and 2 are at the same elevation and connected through the same fluid.

However, this rule only applies when the fluid is continuous. If there is a solid wall between the two points, or if we must jump from one fluid to another, then we cannot automatically say that the pressures are equal.

6 Atmospheric Pressure, Absolute Pressure, and Gauge Pressure

Before solving examples, we need to understand the difference between **absolute pressure** and **gauge pressure**.

Atmospheric pressure is the pressure exerted by the air around us. At sea level, standard atmospheric pressure is approximately

$$101,325 \text{ Pa} = 101.325 \text{ kPa}. \quad (11)$$

It is also equal to about

$$14.7 \text{ psi} = 1 \text{ atm}. \quad (12)$$

Atmospheric pressure can change depending on altitude and weather conditions. At higher elevations, atmospheric pressure is lower.

Absolute pressure is measured relative to a perfect vacuum.

Gauge pressure is measured relative to atmospheric pressure.

The relationship between them is

$$P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atm}}, \quad (13)$$

or

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atm}}. \quad (14)$$

For example, when we measure tire pressure, we usually use gauge pressure. A tire pressure gauge tells us how much higher the pressure inside the tire is compared to the outside atmospheric pressure.

In English units, absolute pressure is often written as psia, and gauge pressure is written as psig. If a value is written only as psi, it usually means gauge pressure unless stated otherwise.

7 Using Hydrostatic Pressure Correctly

Now let us apply these ideas.

Suppose we have a water reservoir with two sides of different widths. Even if one side is wider than the other, the pressure at two points at the same depth in the same continuous water is the same.

This is because hydrostatic pressure depends on depth, not on the shape or width of the container.

The hydrostatic pressure equation is

$$P = P_0 + \gamma h, \quad (15)$$

where P_0 is the pressure at the reference surface, γ is the specific weight of the fluid, and h is the vertical depth below the reference surface.

So pressure depends on the vertical height of the fluid column above the point, not on the total amount of fluid or the container shape.

8 Free Surface and Pressurized Air

A **free surface** is the interface between two fluids. For example, in an open water tank, the free surface is the interface between water and air.

If the tank is open to the atmosphere, the pressure at the free surface is atmospheric pressure. In gauge pressure, this is usually treated as zero:

$$P_{\text{surface}} = 0 \quad \text{gauge.} \quad (16)$$

If one side of the tank is closed and contains pressurized air, then the pressure at the free surface on that side is not atmospheric pressure. The air pressure pushes down on the water and changes the height of the water surface.

If the difference in water height is Δh , the gauge pressure of the trapped air can be found using

$$P_{\text{air,gauge}} = \gamma \Delta h. \quad (17)$$

This comes directly from the hydrostatic pressure relation.

9 Multiple Fluids and Solid Barriers

When more than one fluid is involved, we must be careful.

If two points are in the same fluid and at the same elevation, their pressures are equal only if the fluid is continuous between those points.

However, if we have to cross into another fluid, we cannot automatically say the pressures are equal, even if the two points are at the same height.

For example, if water is on one side and oil is on another side, the pressure changes differently in each fluid because each fluid has a different specific weight.

Also, if there is a solid wall separating two regions of the same fluid, we cannot automatically say the pressures are equal across the wall. The fluid is not continuous through the solid barrier.

Important Rule

Pressure is the same at the same elevation only in the same continuous fluid.

10 Manometer Example

Now let us consider a manometer problem.

Suppose we have a water tank pressurized by air. A manometer containing oil and mercury is connected to the tank to measure the air pressure inside.

The goal is to find the air pressure inside the tank.

To solve this type of problem, we move through the connected fluids step by step. When moving downward in a fluid, pressure increases. When moving upward in a fluid, pressure decreases.

The general rule is:

Moving downward:

$$P_{\text{new}} = P_{\text{old}} + \gamma h. \quad (18)$$

Moving upward:

$$P_{\text{new}} = P_{\text{old}} - \gamma h. \quad (19)$$

For the manometer, we can start from the outside pressure. If the manometer is open to the atmosphere, then the outside pressure is atmospheric pressure for absolute pressure calculations, or zero for gauge pressure calculations.

As we move through the manometer:

1. Start at the outside pressure.
2. Move downward through the mercury column, so add the mercury pressure term.
3. Move through the oil column, adding or subtracting depending on the direction of movement.
4. Move through the water column, again adding or subtracting depending on whether we move downward or upward.
5. Relate the final pressure to the air pressure inside the tank.

For example, the pressure inside the tank can be found by writing

$$P_{\text{inside}} = P_{\text{outside}} + \gamma_{\text{Hg}} h_{\text{Hg}} + \gamma_{\text{oil}} h_{\text{oil}} - \gamma_{\text{water}} h_{\text{water}}. \quad (20)$$

The exact signs depend on the direction we move through each liquid column.

If we use gauge pressure, then

$$P_{\text{outside}} = 0. \quad (21)$$

If we use absolute pressure, then

$$P_{\text{outside}} = P_{\text{atm}}. \quad (22)$$

So the same manometer can be used to find either gauge pressure or absolute pressure, depending on how we define the outside pressure.

11 Summary

In this lecture, we learned several important ideas about fluid pressure:

- Pressure at a point in a fluid acts equally in all directions. This is Pascal's Law.
- Hydrostatic pressure increases with depth.
- When the z -axis points upward,

$$\frac{\partial P}{\partial z} = -\gamma. \quad (23)$$

- Pressure is the same at the same elevation in the same continuous fluid.
- Gauge pressure is measured relative to atmospheric pressure.
- Absolute pressure is measured relative to a vacuum.
- In manometer problems, add pressure when moving downward and subtract pressure when moving upward.

These ideas are the foundation for solving many hydrostatic pressure problems in fluid mechanics.